01NAEX - Lecture 05
Introduction to the 2^k Factorial Design

Jiri Franc

Czech Technical University
Faculty of Nuclear Sciences and Physical Engineering
Department of Mathematics

Introduction to the 2^k Factorial Design

By a factorial design we mean that in each complete replicate of the experiment all possible combinations of the levels of the factors are investigated (Lecture 3: the general factorial design).

- The 2^k factorial design is a special case of the general factorial design with k factors, all at 2 levels.
- The two levels are usually called low $(-)$ and high $(+)$.
- A complete replicate requires $2 \times 2 \times \cdots \times 2 = 2^k$ runs: the smallest number of runs for studying k factors in a complete factorial.
- The 2^k design is widely used in industrial experimentation, especially in the early stages of experimental work (**factor screening**), when many factors are likely to be investigated.

Introduction to the 2^k Factorial Design

Factors can be

- **quantitative**
  (two values of temperature, pressure, time, . . .)
- **qualitative**
  (two machines, two operators, presence and absence of something)

With two levels we can only fit a straight line in each factor: we assume that the **response is approximately linear over the range of the factor levels chosen**.

Assumptions in this lecture

- the factors are fixed,
- the design is completely randomized,
- the usual normality assumptions are satisfied.

Introduction to the 2^k Factorial Design

The analysis procedure for a 2^k factorial design:

1. Select the factors and their ranges (levels).
2. Formulate the design (replicates, randomization).
3. Estimate the factor effects.
4. Formulate the model (initially the full model).
5. Statistical testing (ANOVA).
6. Refine the model (remove nonsignificant terms, respect hierarchy).
7. Analyze the residuals (graphically).
8. Interpret the results (effect plots, response surface, contour plot).

The 2^2 Design - Chemical Process Example

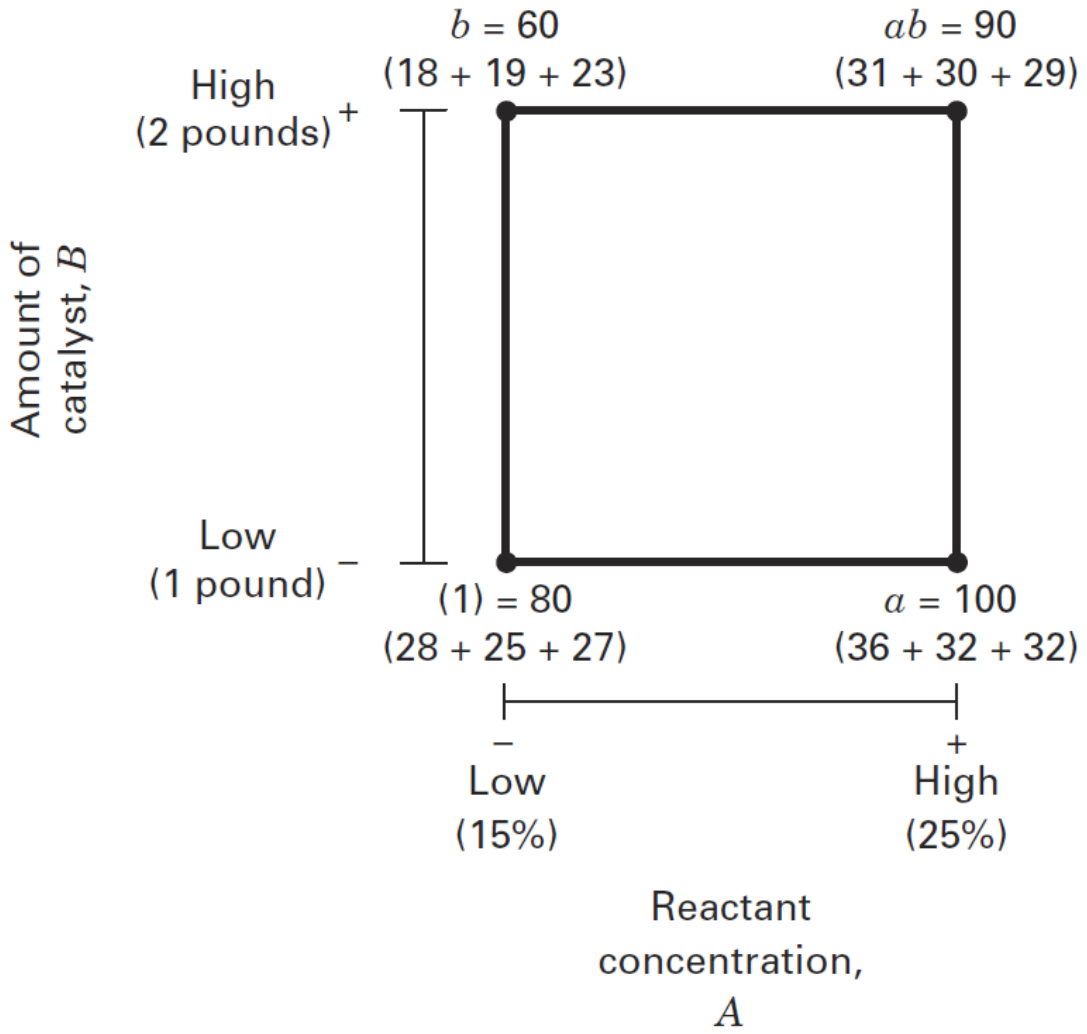
The 2^2 design is the simplest case. We investigate the effect of the concentration of the reactant and the amount of the catalyst on the conversion (yield) in a chemical process (Montgomery, Section 6.2).

	Factor		Treatment	Replicate			
	A	*B*	Combination	I	II	III	Total
(1)	−	−	*A* low, *B* low	28	25	27	80
a	+	−	*A* high, *B* low	36	32	32	100
b	−	+	*A* low, *B* high	18	19	23	60
ab	+	+	*A* high, *B* high	31	30	29	90

- factor A – reactant concentration (15 and 25 %),
- factor B – amount of catalyst (1 and 2 pounds),
- response y – yield (%), $n = 3$ replicates, the 12 runs in random order.

The goal is to determine whether adjustments to these factors would increase the yield.

Treatment Combinations in the 2^2 Design



- Low ($-$) and high ($+$) levels of a factor are arbitrary labels; for a qualitative factor it does not matter which level is called high.
- The high level of a factor in a treatment combination is denoted by the corresponding lowercase letter.
- (1) denotes the combination with all factors at the low level.
- The same symbols (1), *a*, *b*, *ab* also denote the **totals** of the *n* observations at that combination.

The 2^2 Design - Estimation of Factor Effects

The symbols (1), a, b, ab represent the totals of all n replicates taken at the treatment combinations.

main effect of A:	A	$= \frac{1}{2n}[ab + a - b - (1)] = \bar{y}_{A+} - \bar{y}_{A-}$
main effect of B:	B	$= \frac{1}{2n}[ab + b - a - (1)] = \bar{y}_{B+} - \bar{y}_{B-}$
interaction effect AB:	AB	$= \frac{1}{2n}[ab + (1) - a - b]$

The **interaction effect AB** is the average difference between the effect of A at the high level of B and the effect of A at the low level of B:

$$AB = \frac{1}{2}\left[\frac{ab - b}{n} - \frac{a - (1)}{n}\right].$$

The brackets are **contrasts** in the treatment totals, e.g. $\text{Contrast}_A = ab + a - b - (1)$; the signs come from the table of $\pm$ signs (the column of AB is the product of the columns of A and B).

The 2^2 Design - Chemical Process Example

The estimates of the effects in the chemical process example ($n = 3$):

main effect of A	A	$= \frac{1}{2\times 3}(90 + 100 - 60 - 80) = 8.33$
main effect of B	B	$= \frac{1}{2\times 3}(90 + 60 - 100 - 80) = -5.00$
interaction effect AB	AB	$= \frac{1}{2\times 3}(90 + 80 - 100 - 60) = 1.67$

Increasing the concentration increases the yield, increasing the amount of catalyst decreases it; the interaction is small relative to the main effects.

Sums of squares directly from the contrasts: $SS = \frac{(\text{Contrast})^2}{4n}$, here $4n = N = 12$.

$$SS_A = \frac{1}{12}(50)^2 = 208.33$$

$$SS_B = \frac{1}{12}(-30)^2 = 75.00$$

$$SS_{AB} = \frac{1}{12}(10)^2 = 8.33$$

$$SS_T = \sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{n} y_{ijk}^2 - \frac{y_{...}^2}{N} = 323.00$$

$$SS_E = SS_T - SS_A - SS_B - SS_{AB} = 31.33$$

The 2^2 Design - Chemical Process Example in Python

Data `yield.txt`; `yield` is a Python keyword, so we rename the column to `y`. ANOVA of the model with interaction:

```
chem = pd.read_csv(URL + "yield.txt", sep=r"\s+")
chem = chem.rename(columns={"yield": "y"})
m1 = smf.ols("y ~ C(reactant) * C(catalyst)", data=chem).fit()
sm.stats.anova_lm(m1, typ=2)
                          sum_sq  df        F   PR(>F)
C(reactant)               208.33   1  53.191  0.00008
C(catalyst)                75.00   1  19.149  0.00236
C(reactant):C(catalyst)     8.33   1   2.128  0.18278
Residual                   31.33   8
```

Model formulas in `statsmodels` (the same notation as in R):

`y ~ x`	the response `y` is modelled as a function of the predictor `x`
`+ x`, `- x`	add / remove the term `x` (`- 1` removes the intercept)
`x:z`	the interaction of `x` and `z`
`x * z`	`x + z + x:z` (main effects and the interaction)
`C(x)`	treat `x` as categorical (a factor)

The 2^2 Design - Chemical Process Example in Python

The model without interaction and the comparison of the two models:

```
m2 = smf.ols("y ~ C(reactant) + C(catalyst)", data=chem).fit()
sm.stats.anova_lm(m2, typ=2)
              sum_sq  df        F   PR(>F)
C(reactant)  208.33   1  47.269  0.00007
C(catalyst)   75.00   1  17.017  0.00258
Residual      39.67   9

sm.stats.anova_lm(m2, m1)         # F-test: is AB needed?
   df_resid      ssr  df_diff  ss_diff       F  Pr(>F)
0       9.0  39.6667      0.0      NaN     NaN     NaN
1       8.0  31.3333      1.0   8.3333  2.1277  0.1828
```

In a balanced 2^k design the sums of squares do not depend on which other terms are in the model (orthogonality). Removing AB moves its $SS = 8.33$ and its 1 df to the residual.

The 2^2 Design - Regression Model in Coded Variables

Coded variables $x_1, x_2 = \pm 1$: $x_1 = +1$ if the concentration is at the high level and -1 if it is at the low level; x_2 for the catalyst in the same way.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12}\, x_1 x_2 + \varepsilon$$

```
chem["A"] = chem["reactant"].map({"low": -1, "high": 1})
chem["B"] = chem["catalyst"].map({"low": -1, "high": 1})
c1 = smf.ols("y ~ A * B", data=chem).fit()
c1.summary().tables[1]          # confidence interval columns omitted
              coef   std err         t    P>|t|
Intercept   27.5000     0.571   48.135    0.000
A            4.1667     0.571    7.293    0.000
B           -2.5000     0.571   -4.376    0.002
A:B          0.8333     0.571    1.459    0.183
```

$\hat{\sigma} = 1.979$ on 8 df, $R^2 = 0.903$, $F = 24.82$ on 3 and 8 df, $p = 0.0002$.

$$\hat{y} = 27.5 + 4.17\, x_1 - 2.50\, x_2 + 0.83\, x_1 x_2$$

Each coefficient is **one half of the effect** ($\hat{\beta}_1 = 8.33/2$): the effect is the change of $\bar{y}$ over two units of x (from -1 to $+1$). The t-tests are equivalent to the F-tests in the ANOVA ($t^2 = F$).

The 2^2 Design - Reduced Model in Coded Variables

The interaction is not significant ($p = 0.18$), so we remove it from the model:

```
c2 = smf.ols("y ~ A + B", data=chem).fit()
c2.summary().tables[1]
              coef   std err         t    P>|t|
Intercept   27.5000     0.606   45.377    0.000
A            4.1667     0.606    6.875    0.000
B           -2.5000     0.606   -4.125    0.003
```

$\hat{\sigma} = 2.099$ on 9 df, $R^2 = 0.877$, $R^2_{adj} = 0.850$, $F = 32.14$ on 2 and 9 df, $p = 8 \cdot 10^{-5}$.

The estimates of β_0, β_1, β_2 do not change (orthogonal design); only $\hat{\sigma}$, the standard errors and the df change.

$$\hat{y} = 27.5 + 4.17\, x_1 - 2.50\, x_2$$

The 2^2 Design - Regression Model in Natural Units

The same model in the natural units (concentration in %, catalyst in pounds):

```
chem["conc"] = 20 + 5 * chem["A"]          # 15 or 25 %
chem["cat"] = 1.5 + 0.5 * chem["B"]        # 1 or 2 pounds
n2 = smf.ols("y ~ conc + cat", data=chem).fit()
n2.summary().tables[1]
               coef   std err         t     P>|t|
Intercept   18.3333     3.090     5.933     0.000
conc         0.8333     0.121     6.875     0.000
cat         -5.0000     1.212    -4.125     0.003
```

Conversion between coded and natural variables: $x = \dfrac{\text{natural} - (\text{low} + \text{high})/2}{(\text{high} - \text{low})/2}$,

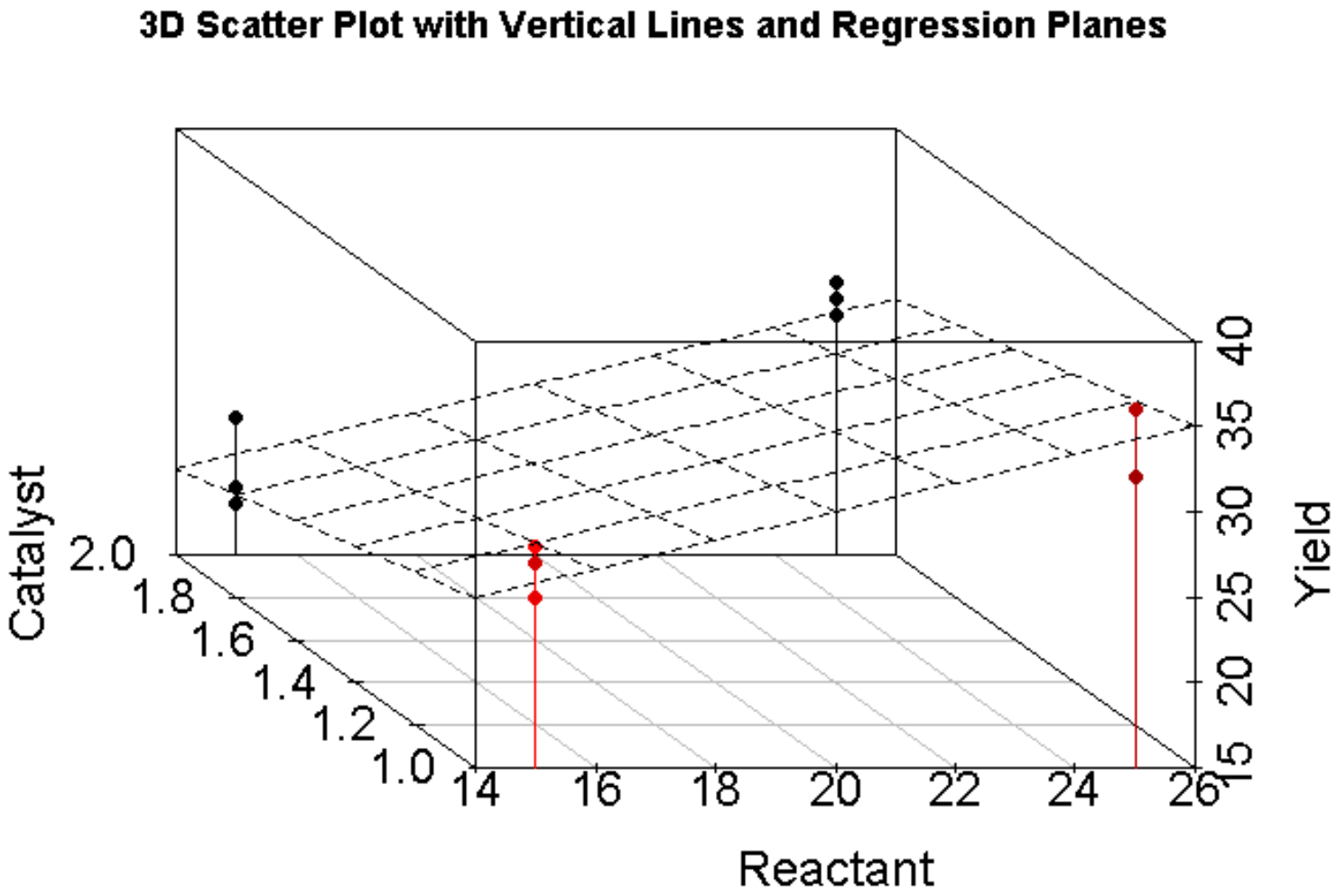
i.e.

$$x_1 = \frac{Conc - 20}{5}, \qquad x_2 = \frac{Cat - 1.5}{0.5},$$

$$\hat{y} = 27.5 + 4.17\,\frac{Conc - 20}{5} - 2.50\,\frac{Cat - 1.5}{0.5} = 18.33 + 0.833\,Conc - 5.00\,Cat.$$

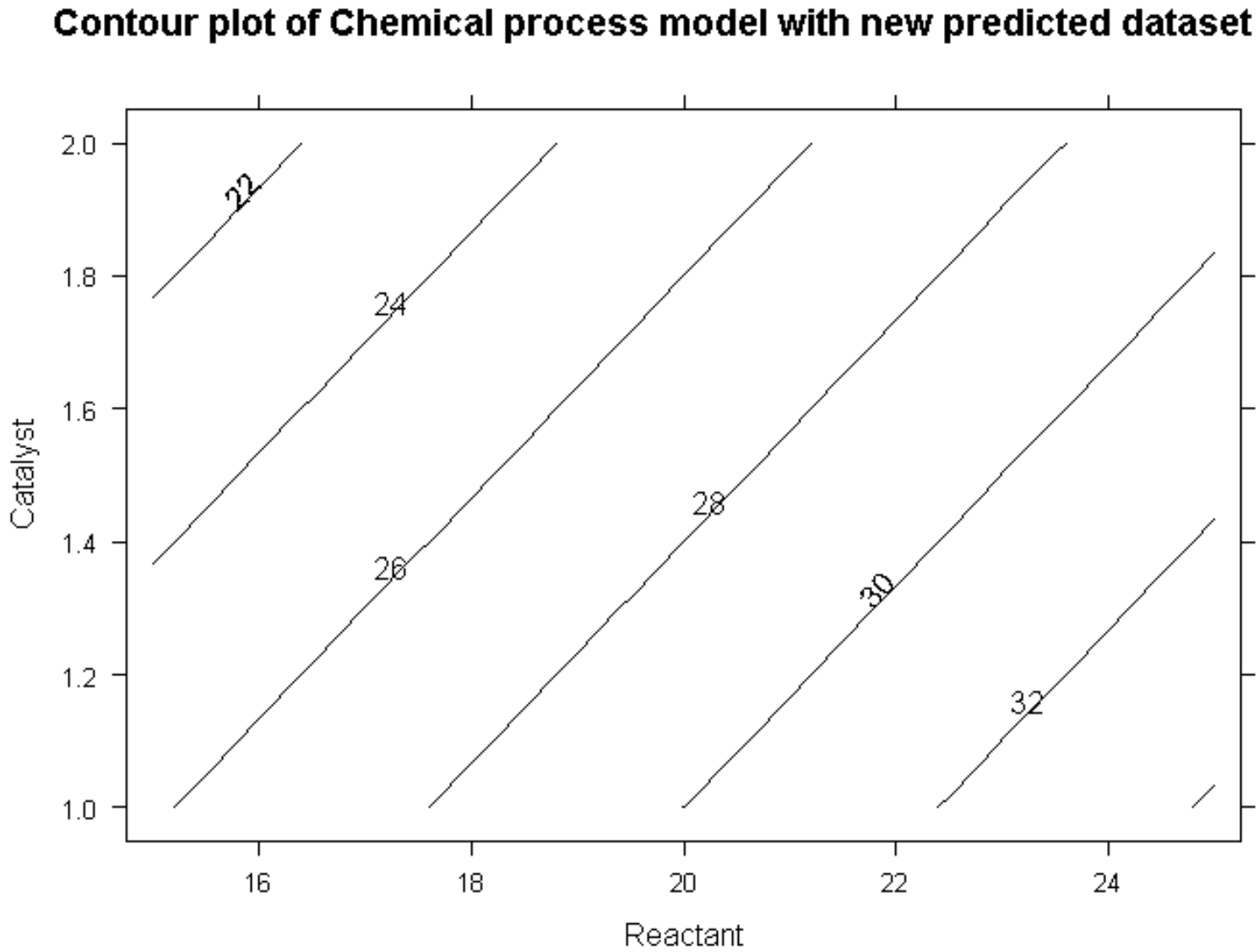
The 2^2 Design - Response Surface

The regression model in natural units can be used to draw the response surface plot.



The 2^2 Design - Contour Plot

. . . and the contour plot. A first-order model without interaction gives a plane, i.e. parallel straight contour lines.



Response Surface and Contour Plot in Python

Predict on a grid of natural values and draw the contours (or a 3D surface):

```
g1, g2 = np.meshgrid(np.linspace(15, 25, 51),
                     np.linspace(1, 2, 51))
grid = pd.DataFrame({"conc": g1.ravel(), "cat": g2.ravel()})
z = n2.predict(grid).to_numpy().reshape(g1.shape)

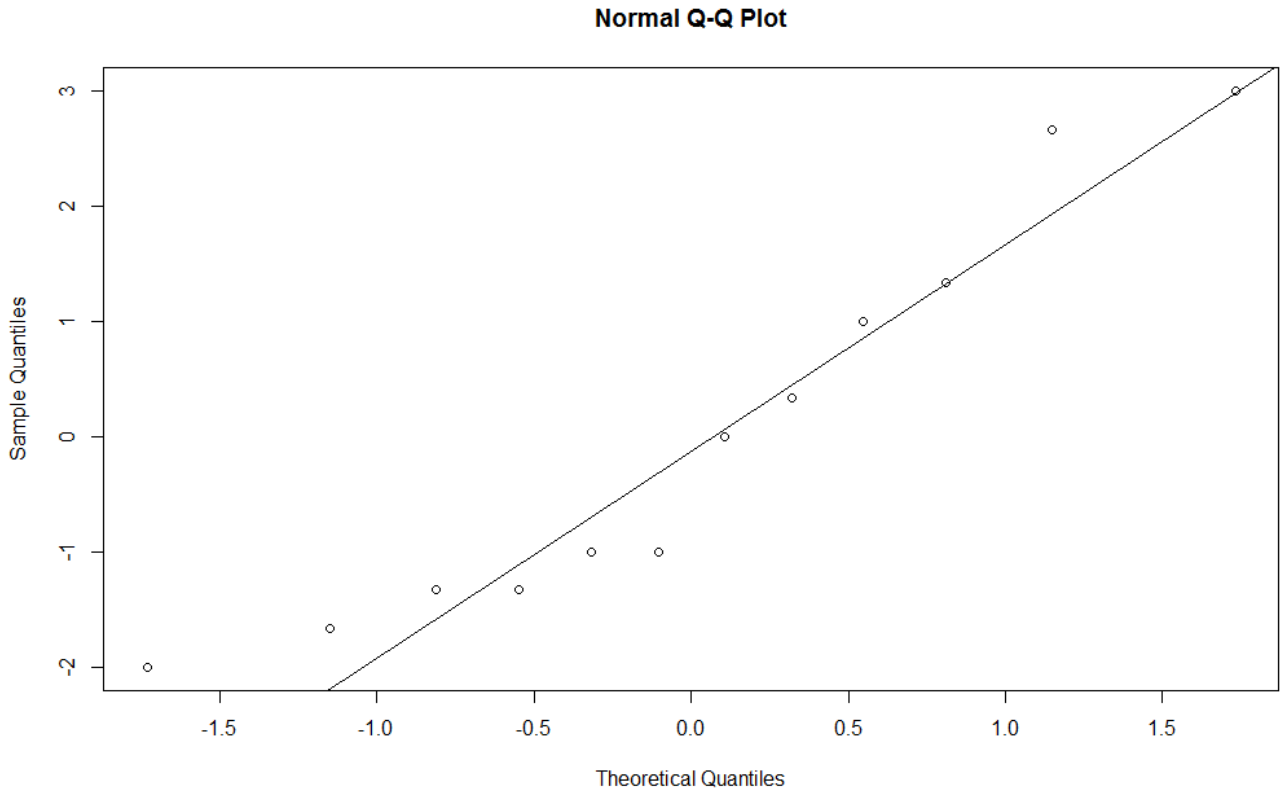
cs = plt.contour(g1, g2, z, levels=10)
plt.clabel(cs)
plt.xlabel("Concentration (%)"); plt.ylabel("Catalyst (pounds)")

ax = plt.figure().add_subplot(projection="3d")
ax.plot_surface(g1, g2, z, cmap="viridis", alpha=0.8)
ax.scatter(chem["conc"], chem["cat"], chem["y"], color="k")
```

Do not extrapolate: the model is supported only inside the region of the experiment ($15 \leq \mathit{Conc} \leq 25$, $1 \leq \mathit{Cat} \leq 2$). With two levels per factor we cannot see curvature (Lecture 7: center points).

The 2^2 Design - Model Adequacy Checking

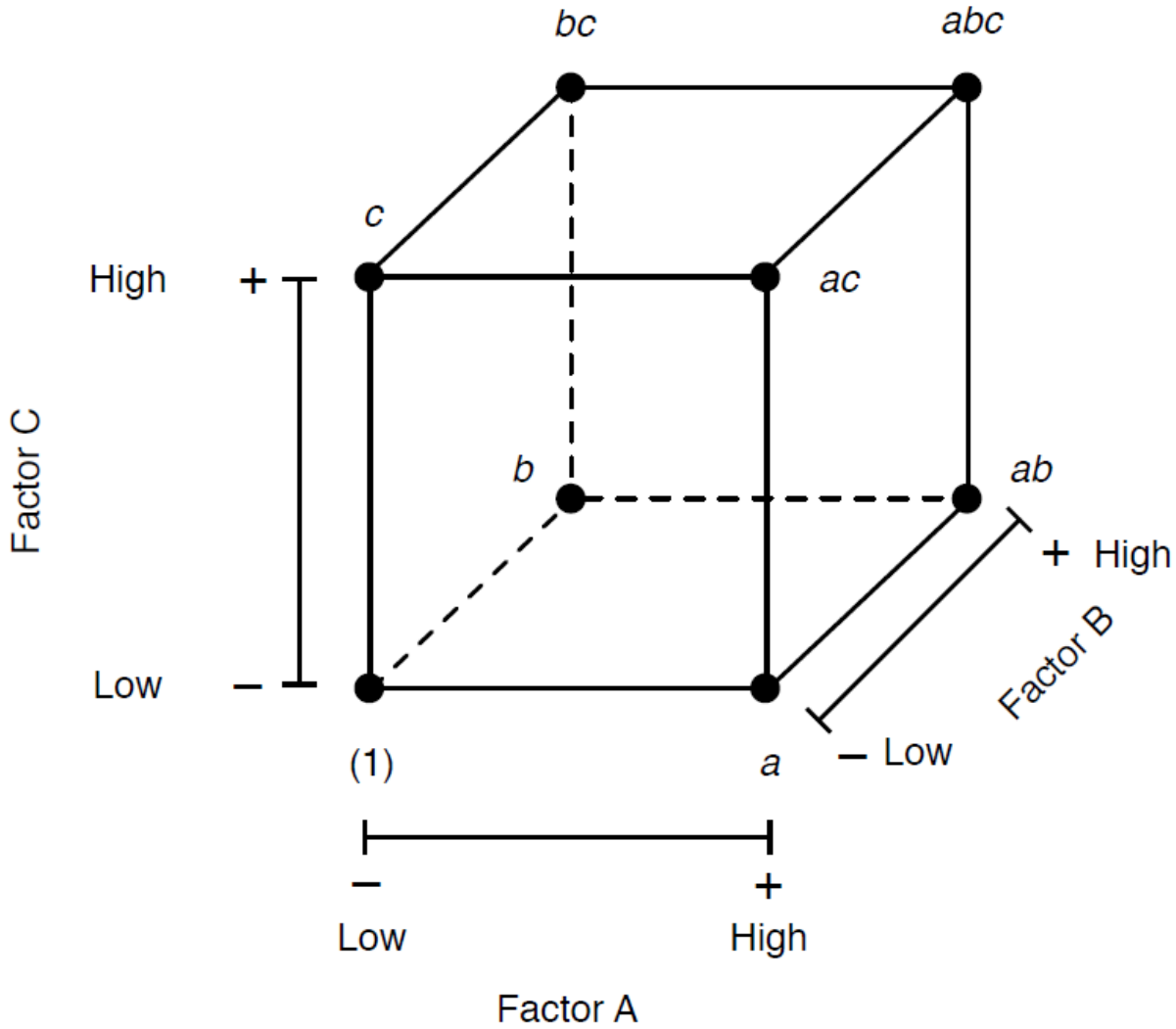
Normal probability plot of the residuals; in Python `sm.qqplot(c2.resid, line="s")`.



Plot also the residuals versus the fitted values, versus each factor and versus the run order; test normality and homoscedasticity.

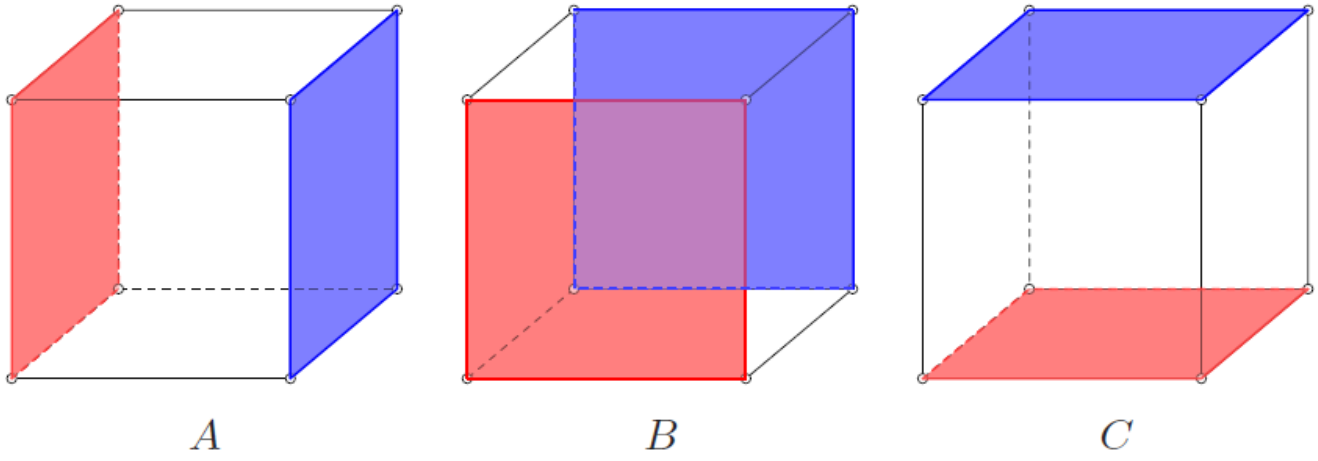
The 2^3 Design

With three factors *A*, *B* and *C* we obtain a 2^3 factorial design with eight treatment combinations that can be displayed as a cube.



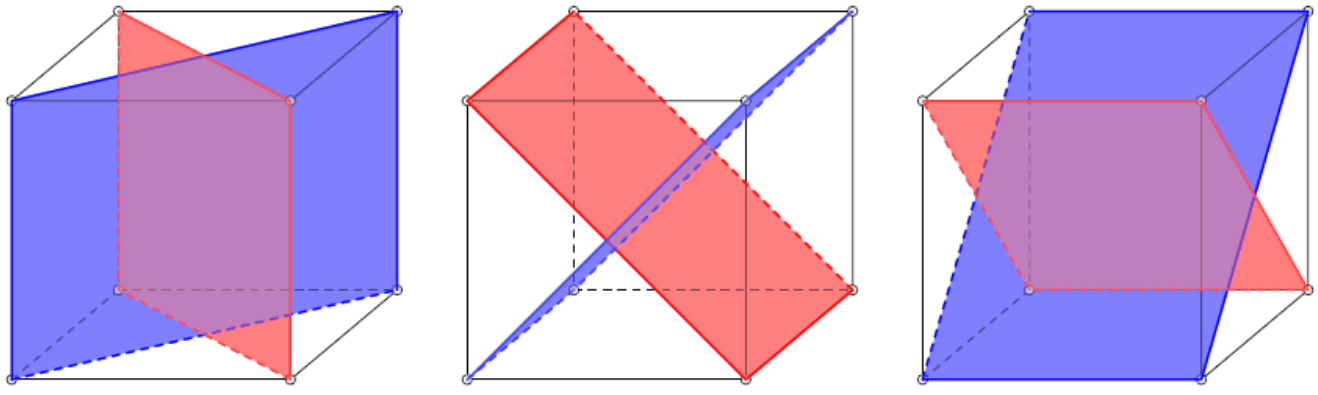
The 2^3 Design

Geometric presentation of the contrasts corresponding to the main effects and interactions in the 2^3 factorial design.



A B C

(a) Main effects

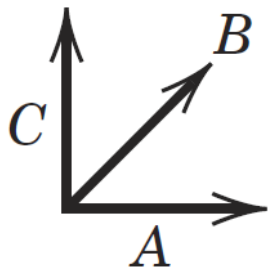


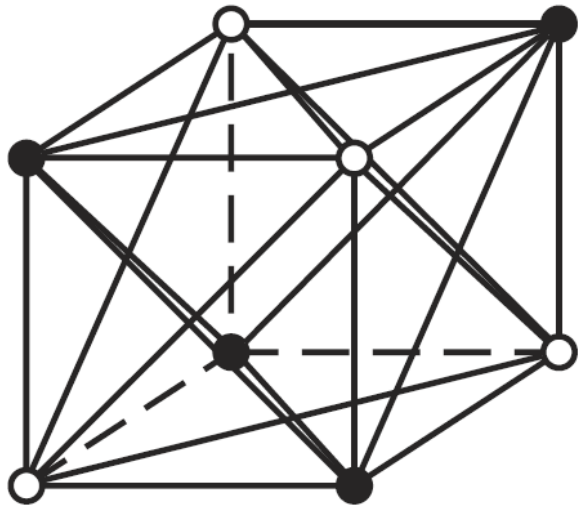
AB AC BC

(b) Two-factor interaction

The 2^3 Design

Geometric presentation of the contrasts corresponding to the main effects and interactions in the 2^3 factorial design.





ABC

● = + runs

○ = − runs

The 2^3 Design - Notation

Three notations used for the runs of a 2^k design:

Run	A	B	C	Label	A	B	C
1	$-$	$-$	$-$	(1)	0	0	0
2	$+$	$-$	$-$	a	1	0	0
3	$-$	$+$	$-$	b	0	1	0
4	$+$	$+$	$-$	ab	1	1	0
5	$-$	$-$	$+$	c	0	0	1
6	$+$	$-$	$+$	ac	1	0	1
7	$-$	$+$	$+$	bc	0	1	1
8	$+$	$+$	$+$	abc	1	1	1

- ± 1 (geometric) notation: the coded variables of the regression model.
- Lowercase letters identify the treatment combinations (and their totals).
- 1 and 0 denote the high and low levels (the notation of Chapter 9).
- The runs are listed in **standard (Yates) order**: A changes fastest, then B, then C. The experiment itself is run in random order.

The 2^3 Design - Table of Plus and Minus Signs

Algebraic signs for calculating the effects in the 2^3 design:

Treatment Combination	Factorial Effect							
	I	A	B	AB	C	AC	BC	ABC
(1)	+	−	−	+	−	+	+	−
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
ab	+	+	+	+	−	−	−	−
c	+	−	−	+	+	−	−	+
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
abc	+	+	+	+	+	+	+	+

- Except for column I, every column has an equal number of $+$ and $-$ signs.
- The sum of products of any two columns is zero: the columns are **orthogonal**.
- The product of any two columns yields a column of the table, e.g. $A \times B = AB$, $AB \times BC = AB^2C = AC$ (since $B^2 = I$).

The 2^3 Design - Estimation of Effects

The estimates of the average effects:

main effect of A:	A	$= \bar{y}_{A+} - \bar{y}_{A-}$
		$= \frac{1}{4n}[a + ab + ac + abc - (1) - b - c - bc]$
main effect of B:	B	$= \bar{y}_{B+} - \bar{y}_{B-}$
		$= \frac{1}{4n}[b + ab + bc + abc - (1) - a - c - ac]$
main effect of C:	C	$= \bar{y}_{C+} - \bar{y}_{C-}$
		$= \frac{1}{4n}[c + ac + bc + abc - (1) - a - b - ab]$
interaction AB:	AB	$= \frac{1}{4n}[abc + ab + c + (1) - a - b - ac - bc]$
interaction AC:	AC	$= \frac{1}{4n}[abc + ac + b + (1) - a - c - ab - bc]$
interaction BC:	BC	$= \frac{1}{4n}[abc + bc + a + (1) - b - c - ab - ac]$
interaction ABC:	ABC	$= \frac{1}{4n}[abc + a + b + c - ab - ac - bc - (1)]$

Sums of squares: $SS = \frac{(\text{Contrast})^2}{8n}$, each with one degree of freedom.

The 2^3 Design - The Plasma Etch Experiment

Montgomery, Example 6.1: etch rate (Å/min) of silicon nitride in a single-wafer plasma etching tool, $n = 2$ replicates. The design factors are

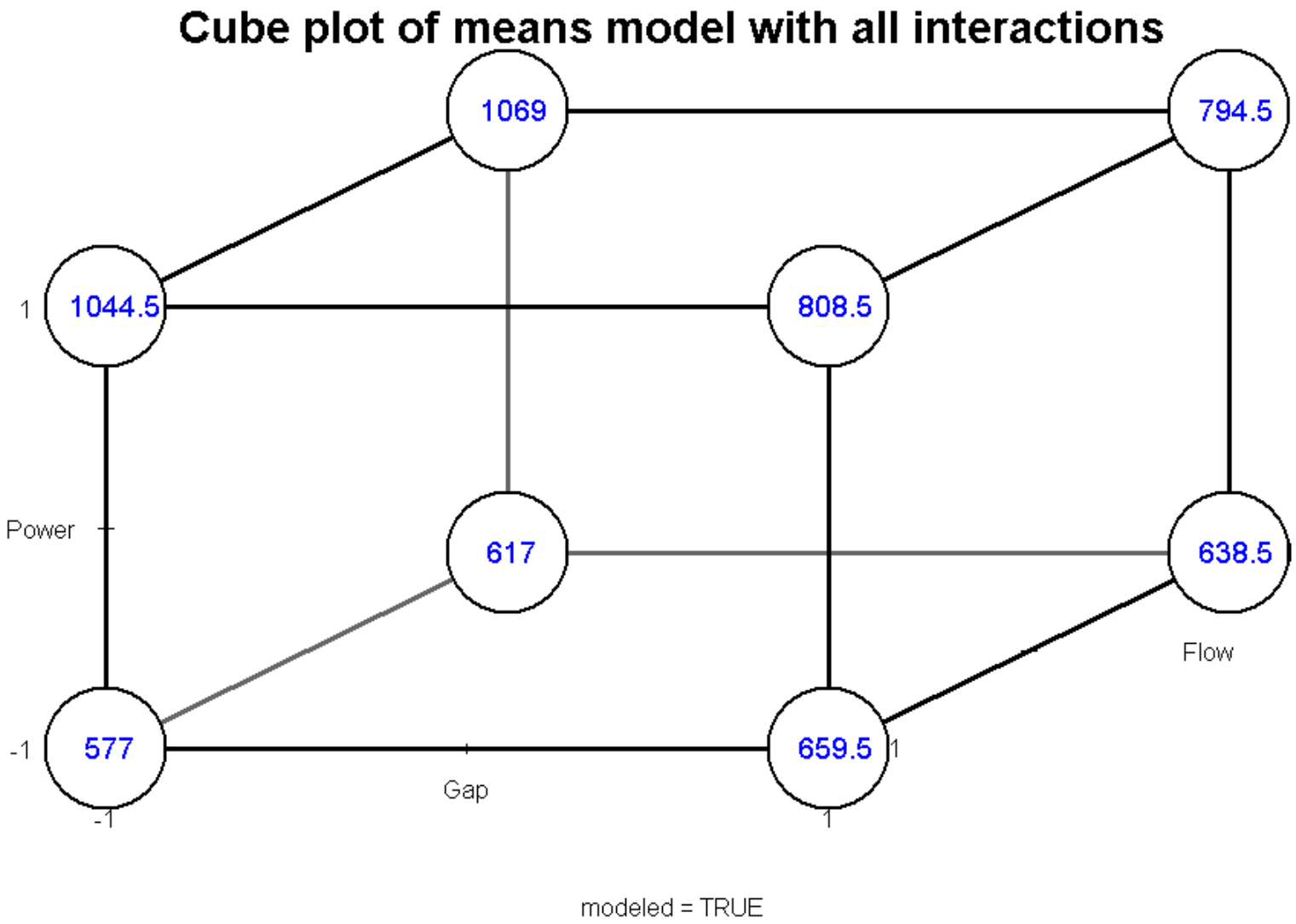
- A: gap between the electrodes (cm), low (-1) 0.8 and high $(+1)$ 1.2,
- B: C_2F_6 gas flow (SCCM), low (-1) 125 and high $(+1)$ 200,
- C: RF power applied to the cathode (W), low (-1) 275 and high $(+1)$ 325.

	Coded Factors			Etch Rate		
Run	A	B	C	Replicate 1	Replicate 2	Total
1	-1	-1	-1	550	604	$(1) = 1154$
2	1	-1	-1	669	650	$a = 1319$
3	-1	1	-1	633	601	$b = 1234$
4	1	1	-1	642	635	$ab = 1277$
5	-1	-1	1	1037	1052	$c = 2089$
6	1	-1	1	749	868	$ac = 1617$
7	-1	1	1	1075	1063	$bc = 2178$
8	1	1	1	729	860	$abc = 1589$

Note: $bc = 1075 + 1063 = 2138$ (the printed total 2178 is a misprint).

The 2^3 Design - The Plasma Etch Experiment

Cube plot of the average etch rates.



Effects from the Table of Signs in Python

Data `plasma.txt` (one row per run, replicates in columns `R1`, `R2`) to the long format; the design matrix of the full coded model is the table of signs.

```
plasma = pd.read_csv(URL + "plasma.txt", sep=r"\s+")
etch = plasma.melt(id_vars=["A", "B", "C"],
                   value_vars=["R1", "R2"], value_name="etch")
full = smf.ols("etch ~ A * B * C", data=etch).fit()
X = full.model.exog[:, 1:]                  # columns A, B, ..., ABC
n, k = 2, 3
contrast = X.T @ etch["etch"].to_numpy()
effect = contrast / (n * 2**(k - 1))
SS = contrast**2 / (n * 2**k)
       contrast   effect           SS
A        -813.0  -101.625   41310.562
B          59.0     7.375     217.562
A:B      -199.0   -24.875    2475.062
C         2449.0   306.125  374850.062
A:C      -1229.0  -153.625   94402.562
B:C        -17.0    -2.125      18.062
A:B:C       45.0     5.625     126.562
```

The largest effects are *C* (power), *AC* and *A* (gap). The effects are exactly $2\hat{\beta}$ of the coded regression model.

The 2^3 Design - The Plasma Etch Experiment

Analysis of variance of the full model:

```
sm.stats.anova_lm(full, typ=2)
              sum_sq  df         F   PR(>F)
A           41310.56   1   18.339  0.00268
B             217.56   1    0.097  0.76391
A:B          2475.06   1    1.099  0.32517
C          374850.06   1  166.411  0.00000
A:C         94402.56   1   41.909  0.00019
B:C            18.06   1    0.008  0.93085
A:B:C         126.56   1    0.056  0.81859
Residual    18020.50   8
```

Gap (A), power (C) and their interaction AC are highly significant; the gas flow B and all interactions with B are not.

$MS_E = 18020.5/8 = 2252.6$ is the **pure error**: it comes only from the differences between the replicates.

We remove the other terms and compare the reduced model with the full one.

The 2^3 Design - Reduced Model and Lack of Fit

Analysis of variance of the reduced model and the test of lack of fit:

```
red = smf.ols("etch ~ A * C", data=etch).fit()
sm.stats.anova_lm(red, typ=2)
              sum_sq  df         F   PR(>F)
A           41310.56   1   23.767  0.00038
C          374850.06   1  215.661  0.00000
A:C         94402.56   1   54.312  0.00001
Residual    20857.75  12

sm.stats.anova_lm(red, full)
   df_resid       ssr  df_diff  ss_diff       F  Pr(>F)
0      12.0  20857.75      0.0      NaN     NaN     NaN
1       8.0  18020.50      4.0  2837.25  0.3149  0.8604
```

The residual of the reduced model = pure error (8 df) + **lack of fit** (the dropped terms B, AB, BC, ABC, $SS = 2837.25$ on 4 df). The lack of fit tested against the pure error: $F_0 = 0.31$, $p = 0.86$, so the reduced model is adequate.

The 2^3 Design - Regression Model in Coded Variables

Full model with coded (± 1) variables; the estimated effects are twice the coefficients.

```
full.summary().tables[1]
              coef   std err         t    P>|t|
Intercept  776.0625    11.865   65.406    0.000
A          -50.8125    11.865   -4.282    0.003
B            3.6875    11.865    0.311    0.764
A:B        -12.4375    11.865   -1.048    0.325
C          153.0625    11.865   12.900    0.000
A:C        -76.8125    11.865   -6.474    0.000
B:C         -1.0625    11.865   -0.090    0.931
A:B:C        2.8125    11.865    0.237    0.819
```

$\hat{\sigma} = 47.46$ on 8 df, $R^2 = 0.966$, $R^2_{adj} = 0.936$, $F = 32.56$ on 7 and 8 df, $p = 2.9 \cdot 10^{-5}$.

All standard errors are equal: the design is orthogonal.

The 2^3 Design - Reduced Model in Coded Variables

The reduced model with coded (± 1) variables:

```
red.summary().tables[1]
               coef   std err         t     P>|t|
Intercept  776.0625    10.423    74.458     0.000
A          -50.8125    10.423    -4.875     0.000
C          153.0625    10.423    14.685     0.000
A:C        -76.8125    10.423    -7.370     0.000
```

$\hat{\sigma} = 41.69$ on 12 df, $R^2 = 0.961$, $R^2_{adj} = 0.951$, $F = 97.91$ on 3 and 12 df, $p = 1.1 \cdot 10^{-8}$.

$$\hat{y} = 776.06 - 50.81\, x_1 + 153.06\, x_3 - 76.81\, x_1 x_3$$

The proportion of the total variability in the etch rate explained by the reduced model is

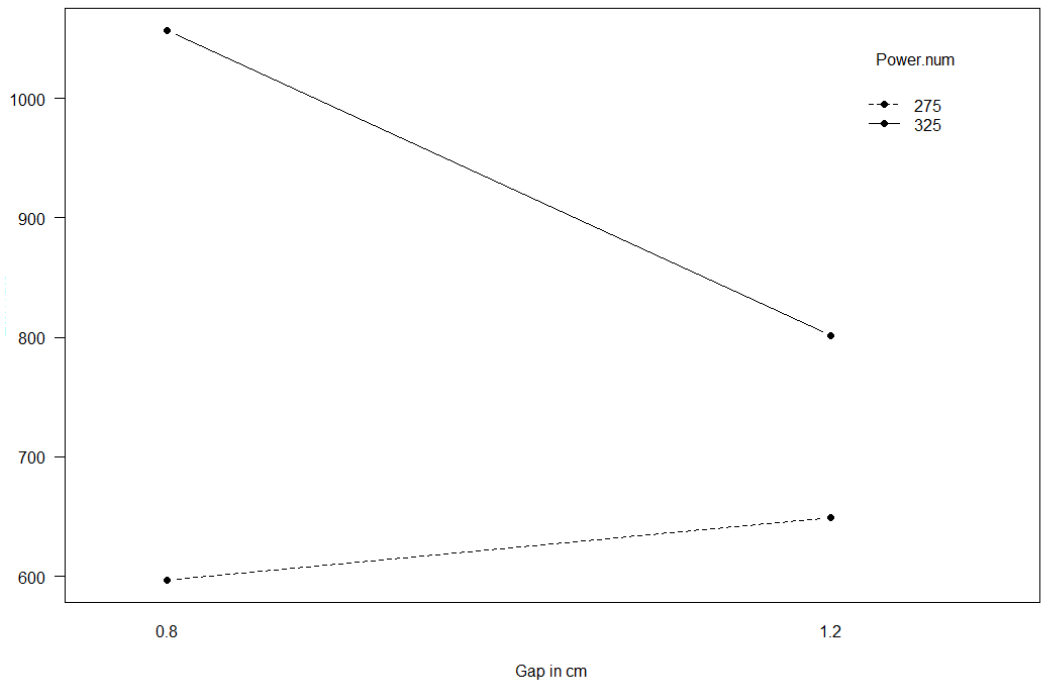
$$R^2 = 1 - \frac{SS_E}{SS_T} = \frac{SS_{Model}}{SS_T} = \frac{SS_A + SS_C + SS_{AC}}{SS_T} = \frac{510563}{531421} = 0.961.$$

R^2 is smaller than for the full model, but R^2_{adj} is larger. Removing the nonsignificant terms gives shorter confidence intervals and better predictions.

The 2^3 Design - Interaction Plot

Interaction of gap and power (averages over *B* and the replicates):

```
from statsmodels.graphics.factorplots import interaction_plot
etch["gap"] = etch["A"].map({-1: 0.8, 1: 1.2})
etch["power"] = etch["C"].map({-1: 275, 1: 325})
interaction_plot(etch["gap"], etch["power"], etch["etch"],
                 markers=["o", "s"], xlabel="Gap (cm)",
                 ylabel="Etch rate")
```



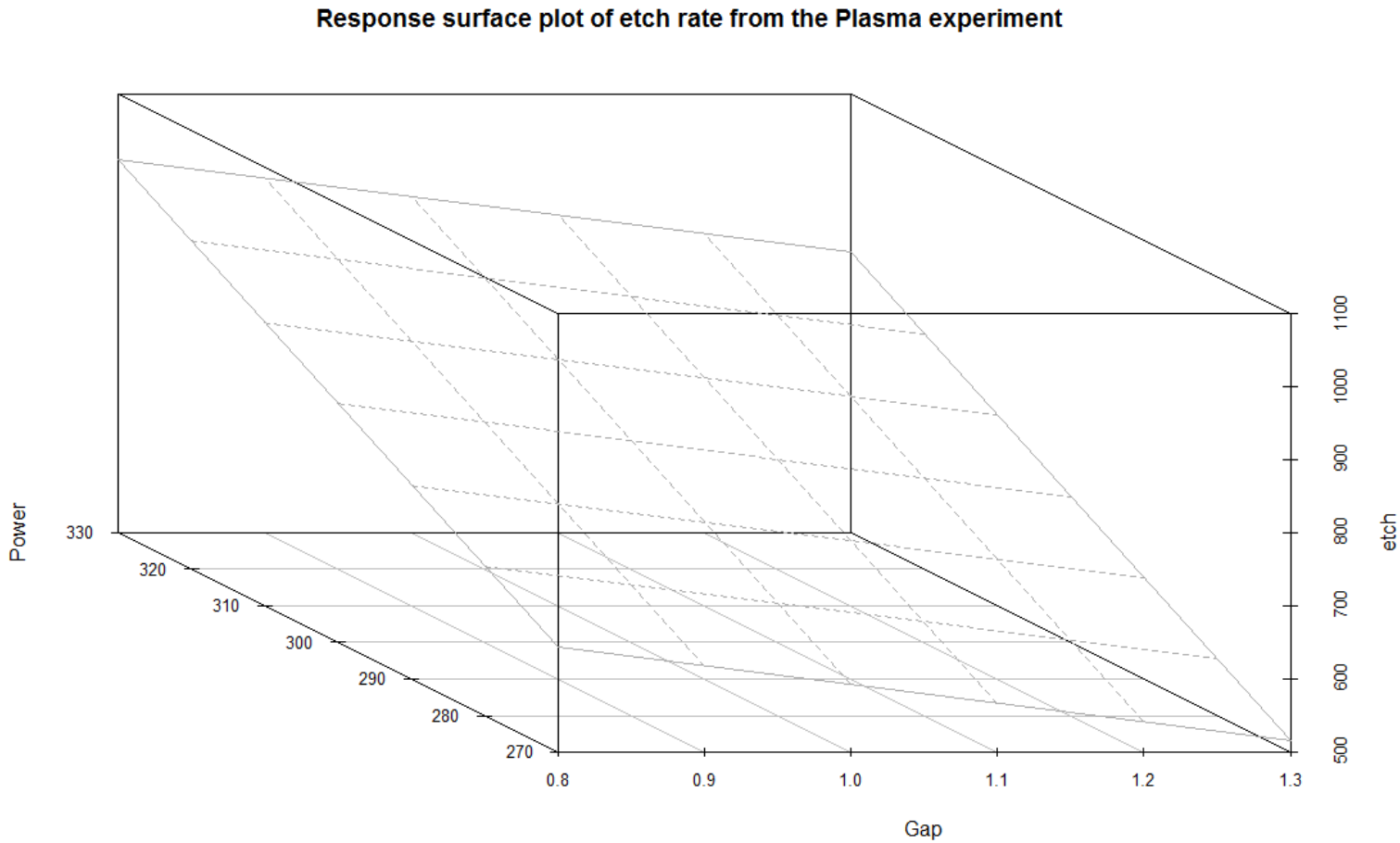
The gap matters a lot at high power and very little at low power.

The 2^3 Design - Response Surface

The fitted reduced model

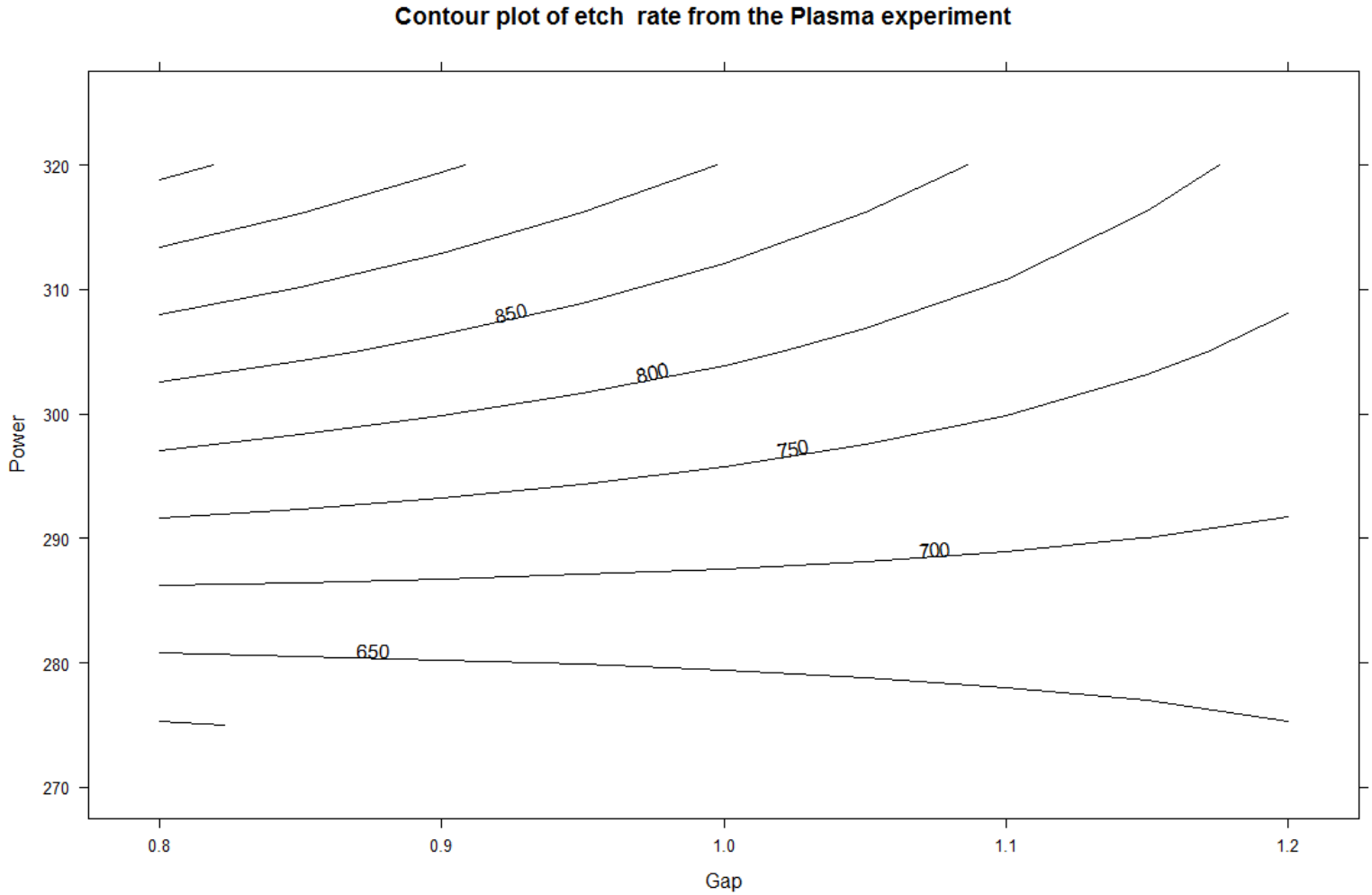
$$\hat{y} = 776.06 - 50.81\, x_1 + 153.06\, x_3 - 76.81\, x_1 x_3,$$

with $x_1 = (Gap - 1.0)/0.2$ and $x_3 = (Power - 300)/25$, drawn in the natural units. The interaction term **twists** the plane.



The 2^3 Design - Contour Plot

The contour lines are curved because of the *AC* interaction.



Standard Errors of the Coefficients and Effects

With n replicates every coefficient and every effect is a contrast of the $N = n2^k$ observations with coefficients $\pm 1/N$ (resp. $\pm 2/N$), so

$$V(\hat{\beta}) = \frac{\sigma^2}{n2^k}, \qquad V(\text{Effect}) = V(2\hat{\beta}) = \frac{\sigma^2}{n2^{k-2}},$$

and the standard errors use $\hat{\sigma}^2 = MS_E$. **All coefficients have the same standard error because the design is orthogonal.**

Plasma etch, reduced model: $se(\hat{\beta}) = \sqrt{1738.1/16} = 41.69/4 = 10.42$.

Confidence interval on a model coefficient:

$$\hat{\beta} - t_{\alpha/2,\, df_E}\, se(\hat{\beta}) \le \beta \le \hat{\beta} + t_{\alpha/2,\, df_E}\, se(\hat{\beta}),$$

$df_E = N - p$ (N runs, p parameters; here $16 - 4 = 12$). For an effect: Effect $\pm$ $t_{\alpha/2,\, df_E}\, se(\text{Effect})$; an interval that excludes zero is the t-test of the coefficient.

Planning: the length $2\, t_{\alpha/2, df_E}\, \sigma/\sqrt{n2^{k-2}}$ decreases with the number of runs $n2^k$; this gives the number of replicates n for a required precision.

Plasma Etch - Confidence Intervals for the Effects

Full model (pure error): $MS_E = 2252.56$ on 8 df, $se(\text{Effect}) = \sqrt{2252.56/(2 \cdot 2^1)} = 23.73$, $t_{0.025,8} = 2.306$, so each interval is Effect $\pm$ 54.72.

Effect	Estimate	95% confidence interval	
A	−101.625	−156.35	−46.90
B	7.375	−47.35	62.10
AB	−24.875	−79.60	29.85
C	306.125	251.40	360.85
AC	−153.625	−208.35	−98.90
BC	−2.125	−56.85	52.60
ABC	5.625	−49.10	60.35

Only the intervals for *A*, *C* and *AC* exclude zero – the same conclusion as the ANOVA.

In Python: `2 * full.conf_int()` (the intervals of the coefficients doubled).

Model Refinement and the Hierarchy Principle

Hierarchy principle: if a model contains an interaction, it should also contain all lower-order terms of the factors involved. E.g. if AC is in the model, keep A and C even if their main effects are not significant.

- A hierarchical model does not depend on the coding of the factors: centering or rescaling a factor (coded $\leftrightarrow$ natural units) gives the same fitted values.
- Without A, the term $x_1 x_3$ alone forces the effect of A to be zero exactly in the middle of the range of C – an assumption we usually cannot justify.
- Hierarchy is a principle, not a law: compare both models (residual mean square, R^2_{adj}, PRESS) – homework Problem 6.44.

Refining the model step by step:

- Start with the full model; with replicates its MS_E is the pure error.
- Remove the nonsignificant higher-order terms first; keep the hierarchy.
- Test the dropped terms together (lack of fit) and check the residuals of the final model.
- In a balanced 2^k design the effect estimates do not change when terms are removed; only MS_E and df_E change.

The General 2^k Factorial Design

A 2^k design has $2^k - 1$ effects (each with 1 df):

- k main effects, $\binom{k}{2}$ two-factor interactions, $\binom{k}{3}$ three-factor interactions, ..., 1 k-factor interaction.

The contrast of the effect $AB \cdots K$ is obtained by expanding

$$\text{Contrast}_{AB\ldots K} = (a \pm 1)(b \pm 1) \cdots (k \pm 1),$$

with the sign $-$ in the bracket of every factor included in the effect and $+$ for the others (then 1 is replaced by (1)). Then

$$\text{Effect} = \frac{2}{n2^k}\,\text{Contrast}, \qquad SS = \frac{1}{n2^k}\,(\text{Contrast})^2, \qquad \hat{\beta} = \frac{\text{Effect}}{2}.$$

Source of variation	Degrees of freedom
k main effects	1 each
$\binom{k}{2}$ two-factor interactions	1 each
...	...
1 k-factor interaction	1
Error	$2^k(n-1)$
Total	$n2^k - 1$

Why We Work with Coded Variables

(Montgomery, Section 6.9)

- The coded coefficients are **dimensionless** and directly comparable: $|\hat{\beta}|$ shows the relative importance of the factors, $2\hat{\beta}$ is the effect.
- The design matrix is orthogonal: all coefficients are estimated independently and with the same (minimum) variance.
- In natural units the coefficients depend on the units and on the location of zero. With an interaction, the coefficient of a main effect is the slope at the *natural zero* of the other factor, far outside the experimental region.

Plasma etch, the same reduced model in natural units:

$$\hat{y} = -5415.4 + 4354.7 \cdot \mathit{Gap} + 21.485 \cdot \mathit{Power} - 15.3625 \cdot \mathit{Gap} \cdot \mathit{Power}.$$

The coefficient of the gap is $+4354.7$ although increasing the gap *decreases* the etch rate in the whole region ($A = -101.6$): it is the slope at $\mathit{Power} = 0\,\text{W}$. The coded model ($\hat{\beta}_1 = -50.81$, slope at the center $\mathit{Power} = 300\,\text{W}$) is the one to interpret; the natural model is for prediction.

Generating a 2^k Design in Python

The full factorial in standard order, *n* replicates and a random run order (fix the seed so that the plan can be reproduced):

```
import itertools
k, n_rep = 3, 2
runs = itertools.product([-1, 1], repeat=k)
std = pd.DataFrame([r[::-1] for r in runs],     # A fastest
                   columns=["A", "B", "C"])
design = pd.concat([std] * n_rep, ignore_index=True)
design["rep"] = np.repeat(np.arange(1, n_rep + 1), 2**k)
rng = np.random.default_rng(2026)
design["run"] = rng.permutation(len(design)) + 1
design.sort_values("run").head(4)
     A  B  C  rep  run
5    1 -1  1    1    1
3    1  1 -1    1    2
4   -1 -1  1    1    3
10  -1  1 -1    2    4
```

Map the codes to the real levels (e.g. `design["A"].map({-1: 0.8, 1: 1.2})`), add an empty column for the response and print the run sheet. If the replicates are run on different days or batches, randomize within each replicate and treat the replicates as blocks (Lecture 7).

Unreplicated 2^k Factorial Design

- A 2^k factorial design with one observation at each corner of the "cube" is called a single replicate (unreplicated) design.
- Very widely used, especially for screening many factors: $2^5 = 32$, $2^6 = 64$ runs.
- If the factor levels are spaced too closely, the noise can overwhelm the signal. More aggressive spacing is usually better.
- Without replication there is a problem with statistical testing:
  - The full model has zero degrees of freedom for error.
  - Pooling high-order interactions to estimate the error (assumes they are negligible).
  - Normal probability plot of the effects (Daniel, 1959), Lenth's method.

Examples and further discussion in the next lecture.

Today Exercise & Next Lecture

Ex05 – `code/01NAEX_Ex05.ipynb` (Python, Google Colab):

- The lecture examples in Python: chemical process (2^2) and plasma etch (2^3) – effects from the table of signs, ANOVA, coded and natural regression models, lack of fit, confidence intervals of effects, response surface and contour plots, residuals.
- Generating a randomized 2^k design.
- Assignment: Problems **6.5, 6.1, 6.2, 6.3, 6.44 and 6.45** (Montgomery, 8th ed.); **6.8** optional.

Assignment: start in class, finish at home as homework. Submit `HW/01NAEX_Ex05_HW_<Surname>.ipynb` by pull request. **Deadline: Sunday 1 November 2026.**

Team Project 1: finish your measurements; deadline Sunday 8 November 2026.

Next lecture (2 November): unreplicated 2^k designs – normal probability plot of the effects, half-normal plot, Lenth's method, transformations (Montgomery, Sections 6.5–6.6).